

Construct a graph representing a TM:

- **Nodes**: states $q \in Q$,
- **Edges**: representing the transition function:
For $((q, a), (q', a', D)) \in \delta$ with
 $q, q' \in Q$, $a, a' \in \Sigma$ and $D \in \{L, R\}$
there is a link from q to q' labelled by
 $a \rightarrow a', D$, and
- **Initial state**: $s \in Q$ indicated
e.g. by an (unlabelled) edge.